

Thus, isotropic material has two independent elastic constants. The constants are usually denoted as

$$C_{1122} = \lambda; \quad C_{1111} = \lambda + 2\mu; \quad C_{1212} = \mu. \quad (41)$$

The constants λ and μ are called *Lamé constants*. Note that, according to the simplified molecular theory of Poisson and Cauchy, isotropic material is described by one constant only. This is equivalent to the condition $\lambda = \mu$ in (41). Many experiments during the nineteenth century proved that, in general, $\lambda \neq \mu$ although for some special isotropic material it could be $\lambda = \mu$. We write (37) by using (41) in the form

$$[C_{ijkl}] = \begin{bmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & \lambda + 2\mu & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix}. \quad (42)$$

With (42) the generalized Hooke's law (3.2-1) for isotropic material becomes

$$\begin{aligned} \sigma_{11} &= \lambda\vartheta + 2\mu E_{11}; & \sigma_{22} &= \lambda\vartheta + 2\mu E_{22}; \\ \sigma_{33} &= \lambda\vartheta + 2\mu E_{33}; & \sigma_{12} &= 2\mu E_{12}; \\ \sigma_{13} &= 2\mu E_{13}; & \sigma_{23} &= 2\mu E_{23}, \end{aligned} \quad (43)$$

where $\vartheta = \text{tr} \mathbf{E} = E_{11} + E_{22} + E_{33}$. Equation (43) can be written as

$$\sigma_{ij} = \lambda\vartheta\delta_{ij} + 2\mu E_{ij}. \quad (44)$$

The generalized Hooke's law for isotropic bodies (44) could be obtained directly from (3.2-1) and the condition that C_{ijkl} is an isotropic fourth-order tensor. It is known that isotropic fourth-order tensors are linear combinations of the following tensors

$$\delta_{ij}\delta_{kl}; \quad \delta_{ik}\delta_{jl}; \quad \delta_{il}\delta_{jk}. \quad (45)$$

Therefore, in general, we have

$$C_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \lambda_1\delta_{ik}\delta_{jl} + \lambda_2\delta_{il}\delta_{jk}, \quad (46)$$

where λ , λ_1 , and λ_2 are constants. However, since C_{ijkl} satisfies $C_{ijkl} = C_{ijlk} = C_{jikl}$ we must have $\lambda_1 = \lambda_2$. Let $\lambda_1 = \lambda_2 = \mu$. Then, (46) simplifies to

$$C_{ijkl} = \lambda\delta_{ij}\delta_{kl} + \mu(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}). \quad (47)$$

By substituting (47) into (3.2-1) we obtain (44). Equation (44) could be easily solved for the components of deformation tensor E_{ij} . Namely, by taking $i = j$ in (44) it follows that

$$\text{tr } \sigma = \Theta = \sigma_{ii} = 3\lambda\vartheta + 2\mu E_{ii} = (3\lambda + 2\mu)\vartheta. \quad (48)$$

From (48) we can determine the first invariant Θ of the stress tensor σ in terms of the first invariant ϑ of the strain tensor $\mathbf{E}$ as

$$\vartheta = \frac{\Theta}{3\lambda + 2\mu}. \quad (49)$$

Also, from (44) it follows

$$E_{ij} = \frac{\sigma_{ij}}{2\mu} - \frac{\lambda\vartheta}{2\mu}\delta_{ij}, \quad (50)$$

so that

$$E_{ij} = \frac{-\lambda\delta_{ij}}{2\mu(3\lambda + 2\mu)}\Theta + \frac{\sigma_{ij}}{2\mu}. \quad (51)$$

The central result of this section is in equations (44) and (51). Both equations are based on the *first* term of series expansion of (3.1-3). In some applications the nonlinear relations between the strain and stress tensor is needed to properly describe the behavior of the material. One way to obtain such a relation (this is the so-called *materially nonlinear theory*) is to use second-order terms in the series expansion of (3.1-3); that is,

$$\sigma_{ij} = C_{ijkl}E_{kl} + \frac{1}{2}D_{ijklrs}E_{kl}E_{rs}, \quad (52)$$

where D_{ijklrs} is a sixth-order tensor. It could be shown that there are 79 independent components of the tensor D_{ijklrs} . Therefore, in general, an elastic body described by (52) has 100 independent material constants. The basic drawback of the theory based on (52) is that it takes into account squares of the gradient of the displacement vector ($\partial u_i / \partial X_j$) through the second term on the right-hand side of (52) and at the same time neglects the same order terms in the expression for the nonlinear strain tensor (since E_{ij} is used in (52) and not $\bar{E}_{ij}$ or $\bar{\bar{E}}_{ij}$).

One way to avoid this inconsistency is to use the stress-strain relation in the form (44) with a nonlinear strain tensor instead of a linear one. Then, with $\bar{E}_{ij}$ instead of E_{ij} we obtain (see Lurie (1980, p. 151))

$$\sigma_{ij} = \lambda(\text{tr } \bar{\bar{E}})\delta_{ij} + 2\mu\bar{\bar{E}}_{ij}. \quad (53)$$

The material described by (53) is called *the Seth* material. Then, the consistent generalization of Hooke's law reads ($\bar{\bar{E}}_{ij}$ instead of E_{ij} in (52))

$$\sigma_{ij} = C_{ijkl}\bar{\bar{E}}_{kl} + \frac{1}{2}D_{ijklrs}\bar{\bar{E}}_{kl}\bar{\bar{E}}_{rs}. \quad (54)$$

Again, there are, in general, 100 independent material parameters in equation (54).